

Deewanasonic Engineering

Artificial Intelligence & Software Development Solutions

Date: March 6, 2026

To: Mr. Benedict Ogazi & Miss Ogazi Chinelotam Stephane (Chinelo)

Institution: Covenant University, Ota, Ogun State

Project Title: Development of an AI-Based Meat Spoilage Detection System

1. Executive Summary & Core Philosophy

At Deewanasonic Engineering, we understand that a final year engineering project is not just about writing code; it is a critical academic milestone. As requested by Mr. Benedict Ogazi, the primary philosophy of this engagement is **Empowerment and Knowledge Transfer**.

Our objective is twofold:

1. To engineer a highly accurate, robust, and factual AI model according to the exact methodology outlined in Chinelo's Chapter 3.
2. To provide step-by-step mentorship so that Chinelo thoroughly understands the underlying Artificial Intelligence (AI) concepts, the Convolutional Neural Network (CNN) architecture, and the codebase, enabling her to confidently defend her project before her university panel.

2. Technical & Educational Deliverables

We will build the system strictly following the **MobileNetV2** architecture and guidelines approved in the project documentation.

Technical Deliverables:

- **Data Processing Pipeline:** Implementation of image resizing (224x224), normalization, and data augmentation using Python and OpenCV.
- **AI Model Development:** Fine-tuning the lightweight MobileNetV2 CNN model using TensorFlow/Keras for binary classification (Fresh vs. Spoiled).
- **Web Application:** Development of a real-time, user-friendly interactive interface using **Streamlit**, allowing users to upload meat images and view predictions.
- **Evaluation Metrics Generation:** Provision of all necessary data visualizations for her Chapter 4 results, including Accuracy, Precision, Recall, F1-Score, Confusion Matrix, and ROC curves.

Educational Deliverables (Knowledge Transfer):

- **Weekly Progress Reports:** Detailed but easy-to-understand updates submitted to both Mr. Ogazi and Chinelo.
- **Code Walkthroughs:** The source code will be heavily commented line-by-line. Virtual sessions will be held to explain the logic to Chinelo.
- **Mock Defense & Q&A Prep:** A final technical grooming session to test her knowledge and prepare her for expected panel questions.

3. Iterative Data Strategy (The "Baseline" Approach)

As discussed, Chinelo has already gathered an initial set of meat images. In professional machine learning, we do not arbitrarily collect massive datasets without testing first.

Our Scientific Approach:

1. We will begin by using Chinelo's existing dataset.
2. We will apply **Data Augmentation** (rotation, flipping, zooming) to artificially expand the dataset and prevent "overfitting" (the AI memorizing data instead of learning).
3. We will train a "Baseline Model" and evaluate its performance.
4. **Decision Point:** If the evaluation metrics (Accuracy, Confusion Matrix) show that the model performs poorly on unseen data, *only then* will we advise and assist in sourcing more data. This teaches Chinelo how to make practical, data-driven engineering decisions.

4. Proposed Timeline (4-Week Schedule)

- **Week 1: Data Preprocessing & The "Baseline" Concept**
 - *Technical:* Cleaning Chinelo's dataset, applying image augmentation, and setting up the Python environment.
 - *Knowledge Transfer:* Explaining to Chinelo how data augmentation works and the concept of overfitting.
 - *Deliverable:* Weekly Report 1.
- **Week 2: Building the CNN Model (MobileNetV2)**
 - *Technical:* Training the AI model on the initial data to recognize fresh vs. spoiled meat.
 - *Knowledge Transfer:* A code walkthrough session breaking down how the MobileNetV2 architecture "sees" and classifies images.
 - *Deliverable:* Weekly Report 2.
- **Week 3: Web Application Integration (Streamlit)**
 - *Technical:* Building the web user interface and connecting it to the trained AI model.
 - *Knowledge Transfer:* Teaching Chinelo how the backend (AI) communicates with the frontend (Web App).

- *Deliverable:* Weekly Report 3.
- **Week 4: Evaluation, Installation & Defense Prep**
 - *Technical:* Generating the Chapter 4 evaluation graphs, bug fixing, and installing the software locally on Chinelo's computer.
 - *Knowledge Transfer:* Teaching her how to interpret her graphs and conducting a "Mock Defense" session.
 - *Deliverable:* Final Handover & Source Code.

5. Project Exclusions (Boundaries)

To ensure academic integrity and align with the approved scope, the following are **strictly excluded** from this service:

1. **Report/Thesis Writing:** Deewanasonic Engineering will provide all the necessary graphs, metrics, and technical explanations, but **Chinelo is solely responsible for writing her Chapter 4, Chapter 5, and the overall project report.**
2. **Hardware/IoT Integration:** As defined in her methodology (Page 17 of her document), the system relies *exclusively* on visual analysis. No physical sensors, microcontrollers, or hardware devices will be built.

Here is the updated **Financials & Payment Structure** section with the 50%, 30%, and 20% milestone split as requested.

6. Financials & Payment Structure

The total cost for the software development, evaluation metrics generation, and the 4-week mentorship program is **₦250,000**.

To ensure mutual commitment and trust, payment will be split into **three strictly defined milestones**:

- **Milestone 1: Project Commencement (50% - ₦125,000)**
 - *Due:* Upon approval of this proposal.
 - *Covers:* Data processing, Baseline Model Training (MobileNetV2), and Weeks 1 & 2 Mentorship/Reports.
- **Milestone 2: App Integration & Evaluation (30% - ₦75,000)**
 - *Due:* Upon successful demonstration/video of the working Streamlit Web App and generation of all Chapter 4 graphs.
 - *Covers:* Web App integration, evaluation metrics generation, and Week 3 Mentorship/Reports.
- **Milestone 3: Final Delivery & Handover (20% - ₦50,000)**
 - *Due:* Prior to the final local installation on Chinelo's PC and final source code handover.
 - *Covers:* System installation on her local machine, Mock Defense prep, and final source code handover.

Note on Additional Costs:

The total cost covers Deewansonic Engineering's software development and mentorship program exclusively. While we aim to use free tiers for platforms like Streamlit wherever possible, any third-party platform costs, hosting subscriptions, unforeseen cloud fees, or external data acquisition costs required for the project will be borne entirely by the client.

Bank Details:

- **Account Name:** Deewansonic Engineering
- **Account Number:** 1025012935
- **Bank:** UBA

7. Next Steps

If the terms of this proposal align with your expectations, kindly process the Milestone 1 payment to the provided account and forward the payment receipt. Work, including the first mentorship scheduling with Chinelo, will commence within 24 hours of confirmation.

We look forward to partnering with you to ensure Chinelo achieves an outstanding grade and a deep understanding of Artificial Intelligence.

Sincerely,

Lead AI Engineer

Deewansonic Engineering